

Power Budget

Team Number: 211

Project Name: EGR 304 Project

Team Member Names: Michael, Levi, Hafsa, Kelton

Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Solenoid	FIT0617	+6- 24V	1	100	100	mA
	Speaker	50CS08FH-1	+5 - 12V	1	625	625	mA
	Transistors	2N904/6	(range)	2	200	200	mA
	MicroController	PIC18F57Q43-CNA	+1.8-5.5V	1	500	500	mA
	+5V Regulator	LM7805CV	+5-35V	1	1000	1000	mA
	MOSFET	STP55NF06	(range)	2	100	100	mA
+9V Power Rail	Component Name	Part Number	Supply	Qty.	Absolute	Total	Unit
	Regulator Input	LM7805CV	+5-35V	1	625	625	mA
	Solenoid	FIT0617	+12 - 24V	1	100	100	mA
	Subtotal					725	mA
	Safety Margin					25%	
	Total Current Required on +9V Rail					906.25	mA

+5V Power Rail	Component Name	Part Number	Supply	Qty.	Absolute	Total	Unit
	PSoC™ 4 BLE Module	PIC18F57Q43-CNAI	+1.8-5.5V	1	200	200	mA
	Speaker	50CS08FH-1	+5 - 12V	1	100	100	mA
	Transistors	2N904/6	(range)	2	200	200	mA
						Subtotal	500
						Safety Margin	25%
						Total Current Required on +5V Rail	625
Regulator	+5V Regulator	LM7805CV	(range)	1	1000	1000	mA

Total Remaining Current Available on +5V Rail						375	mA
Component Name	Part Number	Supply	Output	Absolute	Total	Unit	
Plug-in Wall Supply	(full part number)	9VDC	+V9	5000	5000	mA	
+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA	
+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA	
+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA	
+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA	
Total Remaining Current Available on External Power Source 1					1000		